David S. Rapson

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RESEARCH SPECIALTIES

Energy markets, climate policy, industrial organization, applied microeconomics

ACADEMIC APPOINTMENTS

2021–present	Professor of Economics, UC Davis
2022–2025	Policy Advisor and Assistant Vice President, Federal Reserve Bank of Dallas
2015–2021	Associate Professor of Economics, UC Davis
2008–2015	Assistant Professor of Economics, UC Davis
2016	Visiting Associate Professor, Universidad Carlos III de Madrid, Spain
2012–13	Visiting Researcher, UC Energy Institute, UC Berkeley

PROFESSIONAL LEADERSHIP & AFFILIATIONS

2025–present	Treasurer, Association of Environmental and Resource Economics
2024–present	Co-Editor, Energy Journal
2017–present	Co-Director, UC Davis Energy Economics Program (DEEP)
2025–present	Research Affiliate, Federal Reserve Bank of Dallas
2022–present	Executive Committee, UC Davis Institute of the Environment
2014–present	Editorial Council, Journal of the Association of Environmental and Resource Economics
2021–2022	Treasurer, U.S. Association for Energy Economics
2015–2019	Associate Editor, Journal of Economic Behavior and Organization

EDUCATION

Ph.D., Economics, Boston University, Boston MA, June 2008

M.A., Economics, Queen's University, Kingston ON, Canada, 2003

A.B., Economics, Dartmouth College, Hanover NH, 1999

PUBLICATIONS

"Energy Prices and Electric Vehicle Adoption," with J. Bushnell, E. Muehlegger. *Journal of Political Economy Microeconomics* (forthcoming).

"Smooth Operator? Managing Electric Vehicle Integration in Constrained Distribution Networks," with Blake Shaffer. *Energy Journal* (forthcoming).

"Correcting the Logic Behind the 2026 U.S. Vehicle Emissions Standard Rollback," with A. Bento, K. Gillingham, M. Jacobsen, C. Knittel, B. Leard, J. Linn, J. Sallee, A. van Benthem, K. Whitefoot. *Science* (2026), 393, 869-871.

"The Impact of the 2022 Oil Embargo and Price Cap on Russian Oil Prices," with Lutz Kilian and Burkhard Schipper. *Energy Journal* (2026), 47(3), 1-39.

"Do Bill Shocks Induce Energy Efficiency Investments?" with C. Lang, K. Nakolan, R. Taylor. *Energy Journal* (2025), 46(6), 167-185.

"How to Spend One Trillion Dollars: The U.S. Decarbonization Conundrum," with P. Christensen, P. Baylis, T. Garg, G. McCord, E. Muehlegger, E. Myers, E. Smith. *Nature* (2024), Vol. 634, 1050-1052.

"The Limits and Costs of Full Electrification," with J. Bushnell. *Review of Environmental Economics & Policy*(2024), 18(1), 26-44.

"The End of Neutrality? LCFS, Technology Neutrality, and Stimulating the EV Market," with J. Bushnell, E. Muehlegger, J. Witcover. *Economics of Energy & Environmental Policy* (2024), 13(1).

"Global Transportation Decarbonization," with E. Muehlegger. *Journal of Economic Perspectives* (2023), 37(3): 163-188.

"Economics of Electric Vehicles," with E. Muehlegger. *Review of Environmental Economics & Policy* (2023), 17(2): 274-294.

"Subsidizing Low- and Middle-Income Adoption of Electric Vehicles: Quasi-Experimental Evidence from California," with E. Muehlegger. *Journal of Public Economics* (2022), Volume 216.

"Correcting Estimates of Electric Vehicle Emissions Abatement: Implications for Climate Policy," with E. Muehlegger. *Journal of the Association of Environmental and Resource Economics* (2022), 1(10).

"Low Energy: Estimating Electric Vehicle Electricity Use," with F. Burlig, J. Bushnell and C. Wolfram. *AEA: Papers & Proceedings* (2021), 111, 430-435.

"Estimating the Elasticity to Real Time Pricing: Evidence from the Spanish Electricity Market," with N. Fabra and M. Reguant. *AEA: Papers & Proceedings* (2021), 111, 425-429.

"Future Paths of Electric Vehicle Adoption in the United States: Predictable Determinants, Obstacles and Opportunities," with J. Archsmith and E. Muehlegger. *Environmental and Energy Policy and the Economy* (2021).

"Machine Learning from Schools about Energy Efficiency," with F. Burlig, C. Knittel, M. Reguant and C. Wolfram. *Journal of the Association of Environmental and Resource Economics* (2020). 7(6), 1181-1217.

"Attribute Substitution in Household Vehicle Portfolios," with J. Archsmith, K. Gillingham and C. Knittel. *RAND Journal of Economics* (2020), 51(4), 1162-1196.

"Utilities Included? Split Incentives in Commercial Electricity Contracts," with Katrina Jessoe and Maya Papineau. *Energy Journal* (2020), 41(5), 271-303.

"Does Absolution Promote Sin? A Conservationist's Dilemma," with M. Harding. *Environmental and Resource Economics* (2019), https://doi.org/10.1007/s10640-018-0301-5.

"Flawed Analysis of U.S. Auto Fuel Economy Standards," with A. Bento, K. Gillingham, M. Jacobsen, C. Knittel, B. Leard, J. Linn, V. McConnell, J. Sallee, A. van Benthem K. Whitefoot. *Science* (2018), 362, 1119-1121.

"Regression Discontinuity in Time: Considerations for Empirical Applications," with C. Hausman. *Annual Review of Resource Economics* (2018), 10, 533-552.

"The Rebound Effect and Energy Efficiency Policy," with K. Gillingham and G. Wagner. *Review of Environmental Economics and Policy* (2016), 10(1), 68-88.

"Commercial and Industrial Demand Response Under Time-of-Use Electricity Pricing," with K. Jessoe. *Journal of Industrial Economics* (2015), 63(3), 397-421.

"From Cradle to Junkyard: Assessing the Life Cycle Greenhouse Gas Benefits of Electric Vehicles," with J. Archsmith and A. Kendall. *Research in Transportation Economics* (2015), 52, 72-90.

"Knowledge is (Less) Power: Experimental Evidence from Residential Energy Use," with K. Jessoe. *American Economic Review* (2014), 104(4), 1417-1438.

"Towards Understanding the Role of Price in Residential Energy Choices: Evidence from a Natural Experiment," with K. Jessoe and J. B. Smith. *Journal of Economic Behavior and Organization* (2014), 107(A): 191-208.

"Durable Goods and Long-Run Electricity Demand: Evidence from Air Conditioner Purchase Behavior." *Journal of Environmental Economics and Management* (2014), 68(1): 141-160.

"The Rebound Effect is Overplayed," with K. Gillingham, M. Kotchen, and G. Wagner. *Nature* (2013), 493: 475-476.

"Internet and the Efficiency of Decentralized Markets: Evidence from Automobiles," with P. Schiraldi. *Economics Letters* (2013), 121(2): 232-235.

"Split Incentives in Residential Energy Consumption," with K. Gillingham and M. Harding. *Energy Journal* (2012), 33(2): 37-62.

"Does it pay, at the margin, to work and save? Measuring the effective marginal taxes on Americans' labor supply and savings," with L. Kotlikoff. *Tax Policy and the Economy* (2007), NBER volume 21, MIT Press.

BOOK CHAPTERS

"Utilization and Customer Behavior: Smart Choices for the Smart Grid," with K. Jessoe and J. B. Smith in C-C Liu, S. McArthur, S-J Lee, eds, *Smart Grid Handbook*, John Wiley & Sons, Ltd (2016)

"Would the Fair Tax raise or lower marginal and average tax rates?" with L. Kotlikoff in J. Diamond and G. Zodrow, eds., *Fundamental Tax Reform: Issues, Choices, and Implications*, MIT Press (2008).

AWARDS & HONORS

2022-present	Chancellor's Leadership Professor, UC Davis
2015	Thomas Meyer Distinguished Teaching Award, UC Davis Department of Economics

GRANTS

JPAL North America, 2026. "Improving Equitable Electric Vehicle Adoption" (with T. Garg, K. Gillingham, G. McCord and E. Muehlegger). $211,681.

California Air Resources Board, 2023. "Applying Time-Series Forecasting Models to Projections of CA GHG Emissions and Policy Impacts" (co-PI with J. Bushnell and A. Smith). $417,269.

National Science Foundation, 2021. "Electric Vehicles: Economics of the Extensive and Intensive Margins" (co-PI with F. Burlig and J. Bushnell). $600,000.

Abdul Latif Jameel Poverty Action Lab (J-PAL), 2021. "Estimating the Climate Change Mitigation Potential of Shifting Electric Vehicle Load" (co-PI with F. Burlig and J. Bushnell). $75,000.

Peninsula Clean Energy, 2021. "Understanding the Willingness to Enroll in Electric Vehicle Managed Charging Programs." $220,000.

SB1 2019 (co-PI with J. Bushnell & E. Muehlegger). $61,000.

SB1 2018 (co-PI with J. Bushnell). $79,000.

SB1 2017 & 2018 (co-PI with E. Muehlegger). $50,000 & $78,000.

Sloan Foundation, E2e Project, 2017 (co-PI with Corey Lang). $250,000.

Sloan Foundation, 2017 (co-PI with James Bushnell and Duncan Callaway). $412,000.

National Center for Sustainable Transportation and California Department of Transportation, 2017 (co-PI with E. Muehlegger). $99,000.

National Center for Sustainable Transportation and U.S. Department of Transportation, 2017 (co-PI with E. Muehlegger). $31,000.

Banco Bilbao Vizcaya Argentaria (co-PI with N. Fabra (co-PI) and Mar Reguant). $65,000.

California Air Resources Board. Plug-in electric vehicles in the secondary market (with T. Turrentine (co-PI), G. Tal), 2014. $300,000.

UCE3. Vehicle choice and transaction costs, 2013. $17,000.

California Air Resources Board. Modeling vehicle choice and usage (with P. Mokhtarian (co-PI), K. Gillingham, C. Knittel), 2012. $300,000.

UC Davis, Research Investments in the Sciences and Engineering (with K. Levitt, Ho-Chen, George Barnett, J. Bushnell, A. Scaglioni, and Nicole Woolsey-Biggert), 2012. $840,000.

National Science Foundation (with G. Kesidis, K. Levitt, D.J. Miller, J. Rowe, J. Bushnell, A. Scaglioni), 2012. $240,000 (Rapson share).

UCE3 (with K. Jessoe (co-PI)). Smart grid and choice-neutral defaults, 2012. $17,000.

UCE3 (with K. Jessoe (co-PI)). Smart grid and dynamic electricity pricing, 2010. $17,000.

Junior Faculty Research Grant, UC Davis, 2008, 2009, 2010, 2011, 2012, 2013, 2014.

UC Davis Sustainable Transportation Center. Vehicle choice and transaction frictions, 2008. $70,000

Dean's Fellowship, Boston University, 2003–2008

Summer Research Grant, Boston University, 2007

Graduate Scholarship, Queen's University, 2002–2003

KEYNOTES AND TESTIMONY

2024	Taipei Workshop on Energy Economics, Academia Sinica, Taipei, Taiwan (keynote)
2023	Workshop on Energy & Environmental Economics, Madrid, Spain (keynote)
2022	International Conference on Economic Management and Green Development (keynote)
2021	California Public Utilities Commission En Banc: Energy Rates & Costs (testimony)
2020	Mercatur Research Institute on Global Commons & Climate Change (keynote postponed due to COVID-19)

INVITED PRESENTATIONS

2026	Federal Reserve Bank of Dallas "Energy and the Economy" conference, Institute for Regulatory Law and Economics "Distribution Economics and Policy" workshop, NBER Summer Institute, UC Berkeley Energy Camp, U. Maryland ARE (upcoming)
2025	U. Melbourne, AERE Summer Conference, UC Berkeley Energy Camp, Institute for Regulatory Law and Economics (Northwestern), Santa Fe Institute, Georgia Tech, UT Austin, HEC Montreal.
2024	Academia Sinica, Asia Pacific Industrial Organization Conference (Seoul, Korea), Federal Reserve System Energy Conference, AERE Summer Conference, Society of Government Economists.

2023	Federal Reserve System Energy Conference, Energy Transition & Energy Finance Conference (U. Houston Business School), Data Modernization Workshop (California Energy Commission), U. Navarra, Bern Energy & Environmental Economics Workshop (Bern, Switzerland), UC Berkeley Energy Camp.
2022	UC Irvine, Tufts University, European University Institute (online), Federal Reserve Bank of Dallas (seminar), UC Berkeley Energy Camp, NBER Transportation Economics in the 21st Century, Lone Star College "Curious Minds", Federal Reserve System Energy Conference, Federal Reserve Bank of Dallas "Energy Transition" Conference, Federal Reserve Bank of San Francisco Climate Conference, USAEE Conference.
2021	American Economics Association Annual Meeting (2), NBER Conference on Environmental and Energy Policy and the Economy, UC Davis ITS/STEPS, UC Berkeley Energy Camp
2020	American Economics Association Annual Meeting (discussant), Environmental Defense Fund, UC Energy & Environment Group, NBER, UC Irvine (postponed, COVID-19)
2019	Arizona State University, Claremont McKenna College, POWER Conference (discussant), American Economics Association Annual Meeting, Asia-Pacific Industrial Organization Conference, Kyoto University, Osaka University, Academia Sinica (Taipei), NBER Energy Use in Transportation, Association of Environmental and Resource Economics Annual Conference, UC Berkeley Energy Camp.
2018	Georgetown University, Oxford University, Resources for the Future, UC Riverside, Research Institute of Industrial Economics (Stockholm), Copenhagen Business School, World Congress of Energy and Resource Economics (Sponsored Session), NBER Energy Use in Transportation, Asia-Pacific Industrial Organization Conference.
2017	American Economics Association Annual Meeting, NBER Winter EEE, University of Chicago (Harris), London School of Economics, University of Toulouse, Fondazione Eni Enrico Mattei (Milan), Yale University, Stoney Brook University, UC Berkeley Energy Camp, Conference on Behavioral and Experimental Agri-Environmental Research, Association of Environmental and Resource Economics Annual Conference.
2016	American Economics Association Annual Meeting, Universidad Carlos III de Madrid, University of Mannheim, CESifo (Munich), ETH Zurich, Münster Energy Conference, UC Berkeley Energy Camp, NBER Summer EEE
2015	American Economics Association Annual Meeting, University of Toronto, Association of Environmental and Resource Economics Annual Conference (sponsored session), Renewable Electricity Conference (Van Horne Institute, Calgary, Canada)
2014	Energy Institute at Haas POWER Conference on Energy Research and Policy (discussant), University of California Center for Energy and Environmental Economics, Triangle Resource and Environmental Economics Seminar, College of William & Mary, Carleton University
2013	American Economics Association Annual Meeting, University of California San Diego, University of Colorado Boulder, University Calgary, Association of Environmental and Resource Economics Annual Conference, University of California Energy Institute, University of Tennessee (Baker Institute), Energy Institute at Haas POWER Conference on Energy Research and Policy (discussant), Hong Kong University of Science and Technology, Singapore National University
2012	American Economics Association Annual Meeting (discussant), Santa Barbara (Occasional), CU Boulder, University of California Energy Institute, Energy Institute at Haas POWER Conference on Energy Research and Policy, NBER Summer EEE, University of California Berkeley Agricultural and Resource Economics
2011	University of Arizona, International Industrial Organization Conference, Association of Environmental and Resource Economists Annual Meeting, University of California Center for Energy and Environmental Economics, University of California Energy Institute, Stanford Institute for Theoretical Economics, Resources For the Future, Environmental Defense Fund, University of Melbourne, Australia National University, University of New South Wales, University of Auckland, Motu Research Institute
2010	University of California Energy Institute, Behavior Energy and Climate Change, Bureau of Economic Analysis, Dartmouth College, Nova Scotia Ministry of Energy
2009	University of California Energy Institute, International Industrial Organization Conference
2008	Brown University, Federal Reserve Board, Harvard, HEC Montreal, International Industrial Organization Conference, McMaster University, Queen's University, Stanford University, University of California Davis, NBER EEE (discussant)
2006	NBER Tax Policy and the Economy

TEACHING

Principles of Microeconomics (2016–present)

Ph.D. Energy/Industrial Organization (2009-present)

Energy Economics (2012-present)

Industrial Organization (2009-2014)

Intermediate Microeconomics, Boston University (2006)

PH.D. ADVISING AND PLACEMENT

Robert Reaser — California Independent System Operator (CAISO)

Reid Taylor (Chair) — Federal Reserve Bank of Dallas

Shotaro Nakamura — Federal Trade Commission

Benjamin Dawson — California Independent System Operator

Kevin Nakolan — Environmental Protection Agency

James Archsmith (Chair) — University of Maryland

Bo-Young Choi — Korean Institute of International Economic Policy

Chia-Wen Chen — National Taipei University

Tom Blake (Chair) — eBay Laboratory

Ryan Sandler — Federal Trade Commission

Rebbecca Reed-Arthurs — Consulting

UNIVERSITY & DEPARTMENT SERVICE

2023-2024, 2026–present	UCD Faculty Privilege and Academic Personnel Advisers (member)
2024–present	Undergraduate Alumni Relations, UC Davis Economics Department
2021–present	Graduate Admissions Committee, UC Davis Economics Department
2021–2023	Education Policy Committee, UC Davis College of Letters and Science
2016-2021	Placement Director, UCD Economics Department
2017–2019	Academic Affairs Committee, UC Davis Energy Graduate Group
2010, 2013, 2014, 2015, 2017	UCD Economics Department Recruiting Committee
2011-2013	UCD College Assembly (department representative)
2010–2011	Graduate Advisor, Graduate Studies Committee
2009-2010	Placement Committee, UCD Economics Department (chair)
2008-2009	Placement Committee, UCD Economics Department

REFEREE SERVICE

Econometrica, American Economic Review, Quarterly Journal of Economics, Journal of Political Economy, Review of Economic Studies, American Economic Review: Insights, Journal of the European Economics Association, RAND Journal of Economics, American Economic Journal: Applied Economics, American Economic Journal: Economic Policy, Energy Economics, Review of Economics and Statistics, European Economic Review, Journal of Applied Econometrics, Journal of Economic Behavior and Organization, Journal of the Association of Environmental and Resource Economists, Journal of Environmental Economics and Management, Journal of Industrial Economics, International Journal of Industrial Organization, Journal of Public Economics, Journal of Regulatory Economics, Management Science, National Tax Journal, Journal of Economics and Management Strategy, Energy Journal, Resource and Energy Economics, Regional Science and Urban Economics, Economic Inquiry, Transportation Research Part A, Annual Review of Environment and Resources, Proceedings of the National Academy of Sciences, Nature Cities, Nature Communications, Nature Energy, Joule, Science, European Commission Grant Program, National Science Foundation Grant Program

PREVIOUS WORK EXPERIENCE

Consultant, Cambridge Strategic Management Group (2000–2002)

Consultant, Gen3 Partners, (1999–2000)

Consultant, Mercer Management Consulting (1998)

Citizenship: Canada and United States